

21 Aug '18

AGENDA: 1. PORTFOLIO THEORY / OPTIMIZATION (CAPM)

2. CAPITAL ASSET PRICING MODEL

3. FAMA-FRENCH MODEL

* C++ QUESTIONS

* LINEAR REGRESSION, PCA

PORTFOLIO THEORY.

2 STOCKS - A, B

Returns - r_A, r_B

Var - σ_A^2, σ_B^2

$w_A \rightarrow A$

$w_B \rightarrow B$

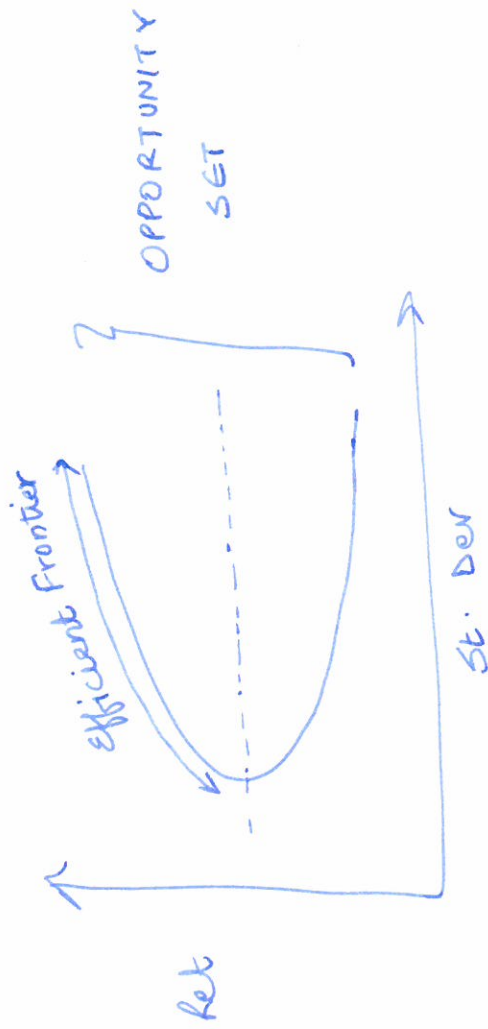
$$w_A + w_B = 1$$

$$\begin{aligned} E[P] &= E[w_A A + w_B B] \\ &= w_A r_A + \cancel{w_B r_B} w_B r_B \end{aligned}$$

$$\begin{aligned} V[P] &= V[w_A A + w_B B] \\ &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \text{Cov}(A, B) \end{aligned}$$

GOAL: MAXIMIZE RETURN, MINIMIZE VARIANCE

Risk Premium: $E[R_p - r_f]$ r_f : Risk-free rate



* ASSUMPTIONS:

1. No TRANSACTION COSTS
2. No TAXES

r_f , $S \rightarrow$ any portfolio on efficient frontier

r_f :- risk-free rate

$$\sigma(r_f) = 0$$

weight x in portfolio S (EF)

$$\text{Bond} : 1 - x$$

$$r_p = x r_s + (1 - x) r_f$$

$$r_p = r_f + x (r_s - r_f)$$

$$E[r_p] = r_f + x E[r_s - r_f] \quad - (i)$$

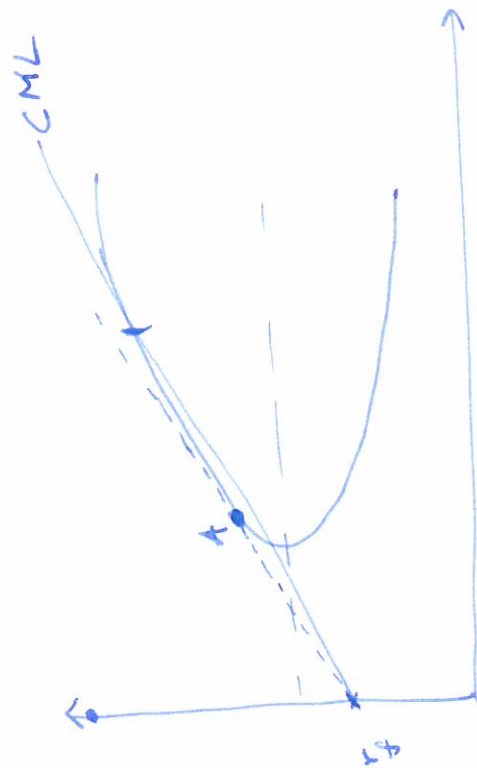
$$V(r_p) = x^2 V(r_s)$$

$$\sigma_p^2 = x^2 \sigma_s^2$$

$$\Rightarrow x = \frac{\sigma_p}{\sigma_s}$$

$$(ii) \Rightarrow E[r_p] = r_f + \sigma_p \times \frac{E[r_s - r_f]}{\sigma_s} \quad - \text{CML [Capital Market line]}$$

$$\text{SHARPE RATIO: } \frac{E[R_S - r_f] \times \sqrt{P}}{\sqrt{S}}$$



A:- TANGENCY PORTFOLIO:- Max^m Sharpe Ratio on EF

PORTFOLIO OF N STOCKS:

N stocks

Returns :- $r_1, r_2, \dots, r_N \rightarrow \bar{R} = [r_1, r_2, \dots, r_N]^T$

↓ same

$\alpha_1, \alpha_2, \dots, \alpha_N \rightarrow \bar{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_N]^T$

Weights :- $w_1, w_2, \dots, w_N \rightarrow \bar{w} = [w_1, w_2, \dots, w_N]^T$

$$R_p = \sum_{i=1}^N w_i r_i$$

$$R_p = \bar{w}^T \bar{R}$$

$$\sigma_p^2 = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \text{Cov}(r_i, r_j)$$

$$\sigma_p^2 = \bar{w}^T \Sigma \bar{w}$$

$\Sigma \rightarrow$ Covariance Matrix

CONSTRUCTING AN EFFICIENT FRONTIER

OPTIMIZATION PROBLEM:-

$$\min \frac{1}{2} \sum_i w_i w_j \text{cov}(r_i, r_j) \quad \text{or} \quad \min \frac{1}{2} \bar{w}^T \Sigma \bar{w}$$

subject to constraints:-

$$1. \sum_{i=1}^N w_i = 1 \quad \text{or} \quad \bar{w}^T \mathbf{1}_N = 1$$

$$2. \sum_{i=1}^N w_i^2 r_i = r \quad \text{or} \quad \bar{w}^T \bar{r} = r \quad | \quad \text{OR} \quad \bar{w}^T \alpha = \alpha_0$$

LAGRANGE MULTIPLIER

$$L(w, \lambda_1, \lambda_2) = \frac{1}{2} \bar{w}^T \Sigma \bar{w} + \lambda_1 (\alpha_0 - \bar{w}^T \alpha) + \lambda_2 (1 - \bar{w}^T \mathbf{1}_N)$$

Differentiate:

$$\frac{dL}{dw} = 0 = \Sigma \bar{w} - \lambda_1 \alpha - \lambda_2 \mathbf{1}_N$$

$$\frac{dL}{d\lambda_1} = 0 = \alpha_0 - \omega^T \alpha$$

$$\frac{dL}{d\lambda_2} = 0 = 1 - \omega^T 1_N$$

$$\boxed{\omega_0 = \lambda_1 \Sigma^{-1} \alpha + \lambda_2 \Sigma^{-1} 1_N}$$

Solve for λ_1 and λ_2

$$\alpha_0 = \omega^T \alpha = \lambda_1 (\alpha^T \Sigma^{-1} \alpha) + \lambda_2 (\alpha^T \Sigma^{-1} 1_N)$$

$$1 = \omega_0^T 1_N = \lambda_1 (\alpha^T \Sigma^{-1} 1_N) + \lambda_2 (1_N^T \Sigma^{-1} 1_N)$$

$$\begin{bmatrix} \alpha_0 \\ 1 \end{bmatrix} = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}$$

$$a = \alpha^T \Sigma^{-1} \alpha, \quad b = \alpha^T \Sigma^{-1} 1_N$$

$$c = 1_N^T \Sigma^{-1} 1_N$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}^{-1} \begin{bmatrix} x_0 \\ 1 \end{bmatrix}$$

QUESTIONS:

1. IDIOSYNCRATIC RISK: UNSYSTEMATIC RISK:- Variance which can be reduced by diversification. Little or no correlation with market risk, thus, can be mitigated.
2. SYSTEMATIC RISK: Variance / Risk which can't be diversified inherent to the market. 'UNDIVERSIFIABLE RISK'. 'MARKET RISK'.

3. PORTFOLIO REBALANCING: Realigning the weights of a portfolio assets by buying or selling assets in portfolio to maintain desired level of allocation.

CAPM

Describes relationship between systematic risk and expected return of assets.

→ Market risk premium

$$E[r_i] = r_f + \beta_i [E[r_m] - r_f] \quad \text{--- SECURITY MARKET LINE (SML)}$$

$\downarrow$ Asset i's return $\downarrow$ Market Return

Beta

(Measure of systematic risk)

$$\beta_i = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)}$$

Asset 'A' $r_A = 8\%$ $\rightarrow$ Actual return

$$\beta = 0.8$$

$$r_f = 3\%$$

$$E(r_m) - r_f = 5\%$$

Use CAPM

$$E[r_A] = 3 + 0.8 \times 5 = \underline{\underline{7\%}}$$

Actual Return above SML, BUY the stock.

FAMA - FRENCH MODEL - 3 FACTOR MODEL

Extension on CAPM

① MARKET

② SIZE [small outperforming large]

③ VALUE [high book to market ratio]

$$E[r_i] = \alpha_f + \beta_m [E(r_m) - r_f] + \beta_s (SMB) + \beta_v (HML) + \epsilon_{error}$$

* LINEAR REGRESSION to find β 's.

* MOMENTUM

* 'INVESTING WITH STYLE' - AQR Capital

C++ QUESTIONS:

1. Features of Object Oriented Programming:

- Abstraction
- Polymorphism
- Inheritance
- Encapsulation

2. What are static variables?

3. What are virtual functions?

4. What are friend functions?

5. What is Run-Time Polymorphism?

6. Abstract Classes? (ABSTRACT CLASSES)

7. Modes of Inheritance?

LINEAR REGRESSION:

1. ASSUMPTIONS OF OLS (Ordinary Least Squares)
2. WEIGHTED LEAST SQUARES Regression
3. Derivations, Equations, Formulae
4. How do you overcome problem of autocorrelation
5. Goodness of fit characteristics

PRINCIPAL COMPONENT ANALYSIS (PCA)

1. Describe PCA, why is it used?
2. SVD:- Singular Value Decomposition
3. Eigen Value Decomposition
4. Math behind PCA.

* Basic Hypothesis Testing